

Assignment 4 — Customer, Product & Order Analysis

Datasets

Hey, there are these **three separate files**:

```
customers.csv  
products.csv  
orders.csv
```

Objective

You are working as a business data analyst. The company stores customer, product, and order information in separate datasets. Your task is to combine these datasets and perform a complete business analysis. This assignment focuses heavily on **merging and combining DataFrames**.

Part A — Understanding Individual Datasets

1. Load all three CSV files.
2. Display the dimensions of each DataFrame.
3. Identify the columns in each dataset.
4. Identify the common columns between datasets.
5. Determine how the three datasets are related.

Part B — Data Cleaning

6. Check each DataFrame for missing values.
7. Check for duplicate records.
8. Check the validity of Customer_ID and Product_ID.
9. Convert Order_Date into datetime.
10. Check whether every order corresponds to a valid customer.
11. Check whether every order corresponds to a valid product.

Part C — Combining Data

12. Merge orders with customers.
13. Merge the resulting DataFrame with products.
14. Create a final consolidated DataFrame.

The final DataFrame should contain information such as:

```
Order_ID  
Customer_ID  
Customer_Name  
City  
Age  
Gender  
Product_ID  
Product_Name
```

Category
Price
Quantity
Order_Date

Part D — Business Calculations

15. Create:

$$\text{Order_Value} = \text{Price} \times \text{Quantity}$$

16. Calculate total revenue.

17. Calculate average order value.

18. Find total revenue by product category.

19. Find total revenue by city.

20. Find the top 10 customers by spending.

21. Find the top 10 products by revenue.

22. Find the most frequently purchased product.

23. Find the city with the highest number of customers.

24. Find the city generating the highest revenue.

Part E — Customer Analysis

25. Divide customers into age groups.

18–30
31–45
46–60
61+

26. Calculate revenue generated by each age group.

27. Compare spending between male and female customers.

28. Identify high-value customers.

29. Determine the percentage of customers who have placed more than one order.

Part F — Advanced Analysis

30. Create a customer spending ranking.

31. Identify the top 10% of customers based on total spending.

32. Find the most profitable product category for each city.

33. Find the most popular product in each category.

34. Calculate monthly revenue.

35. Identify the month with the highest revenue.